🔌 MCP Protocol & MCP Servers — A Visual Guide

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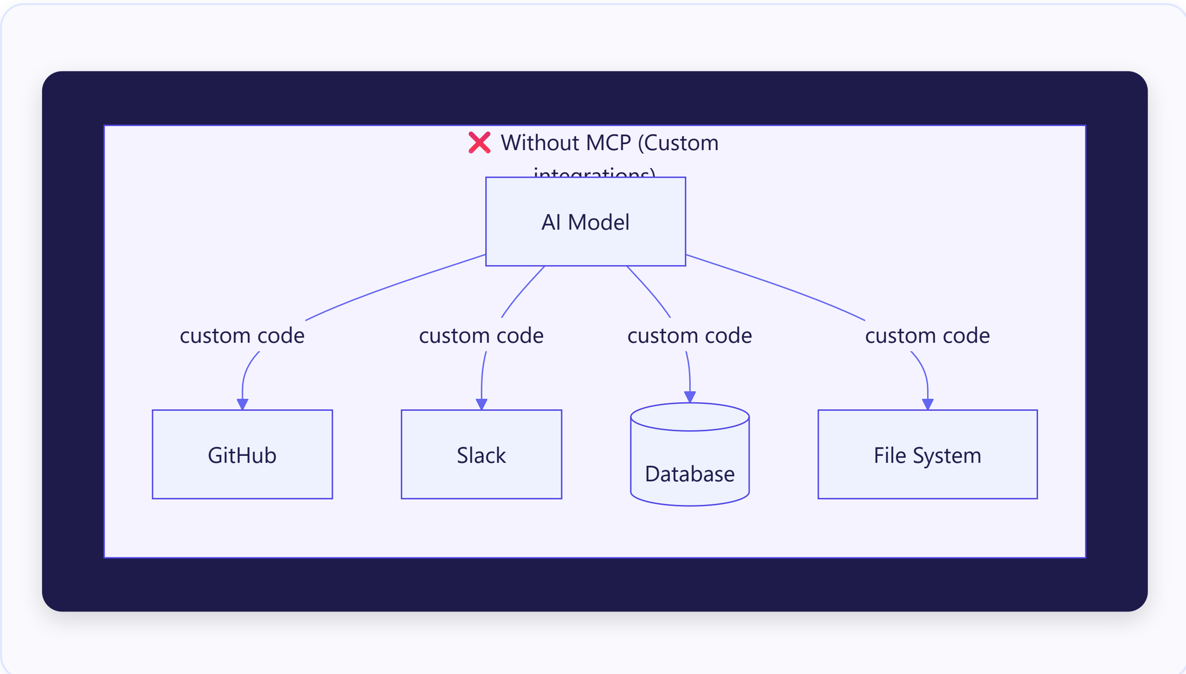
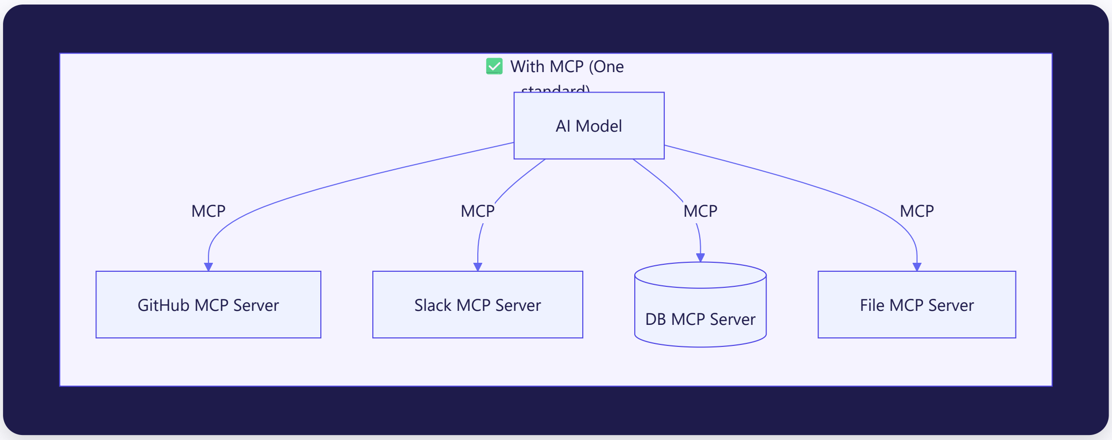
1. What Problem Does MCP Solve?

Before MCP, every AI assistant had to build **custom, one-off integrations** with every tool it wanted to use (like GitHub, Slack, databases, etc.). This was messy and hard to scale.

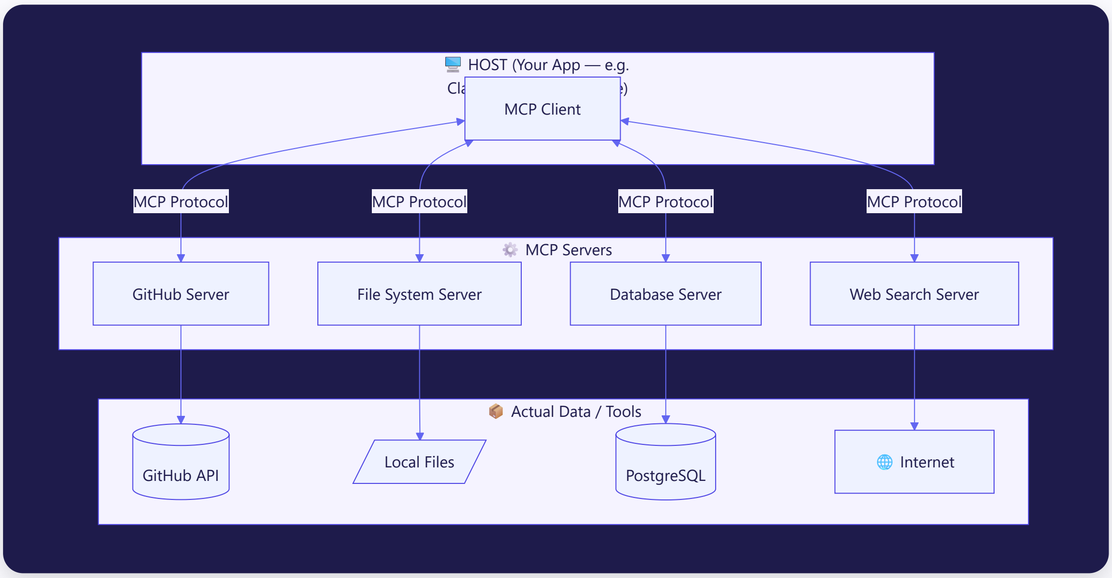
MCP (Model Context Protocol) is a universal standard — like a "USB-C for AI" — that lets any AI model talk to any tool using one consistent protocol.





2. The Big Picture Architecture

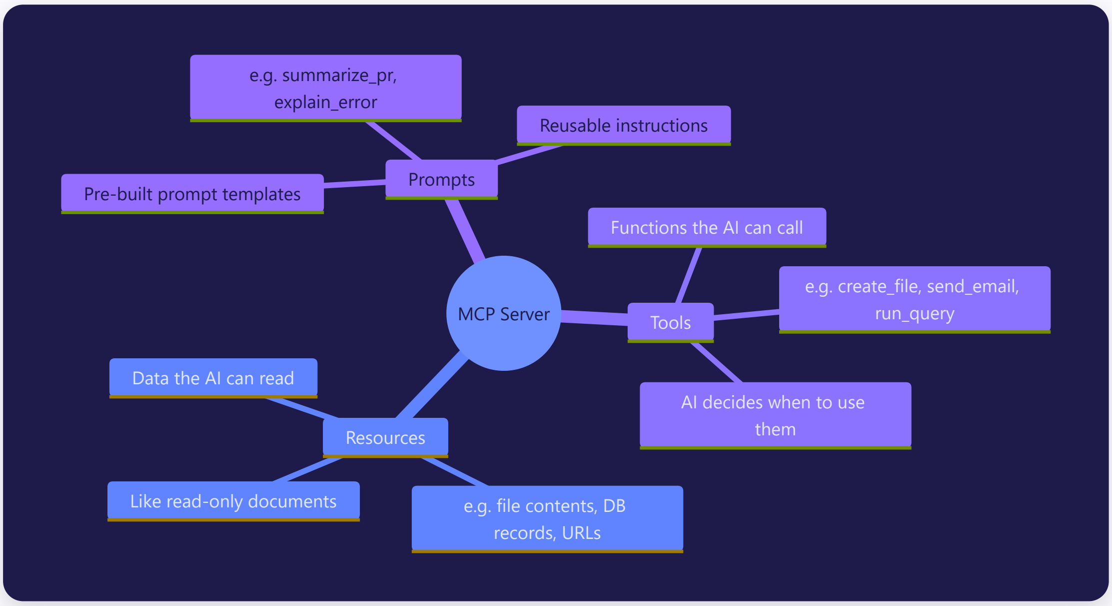
MCP has **3 core players**:



Player	Who They Are	Analogy
Host	The app you're using (Claude Desktop, VS Code)	Your phone
MCP Client	Lives inside the host, speaks MCP	The App Store engine on your phone
MCP Server	A lightweight program exposing a tool/data source	An app you install

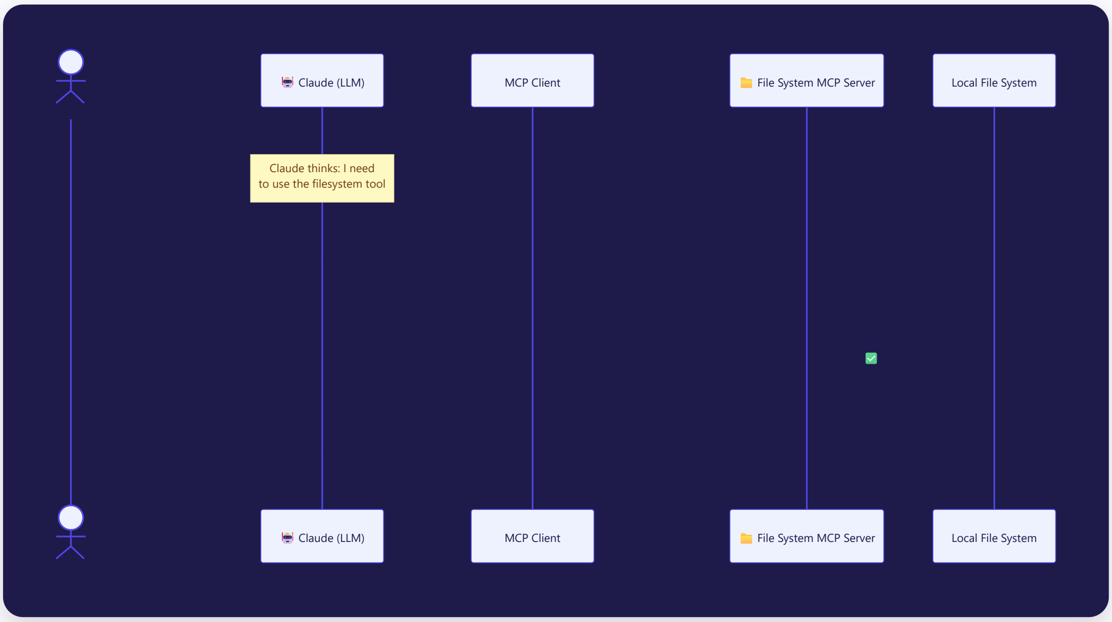
3. What Can an MCP Server Offer?

Each MCP server can expose **3 types of capabilities**:



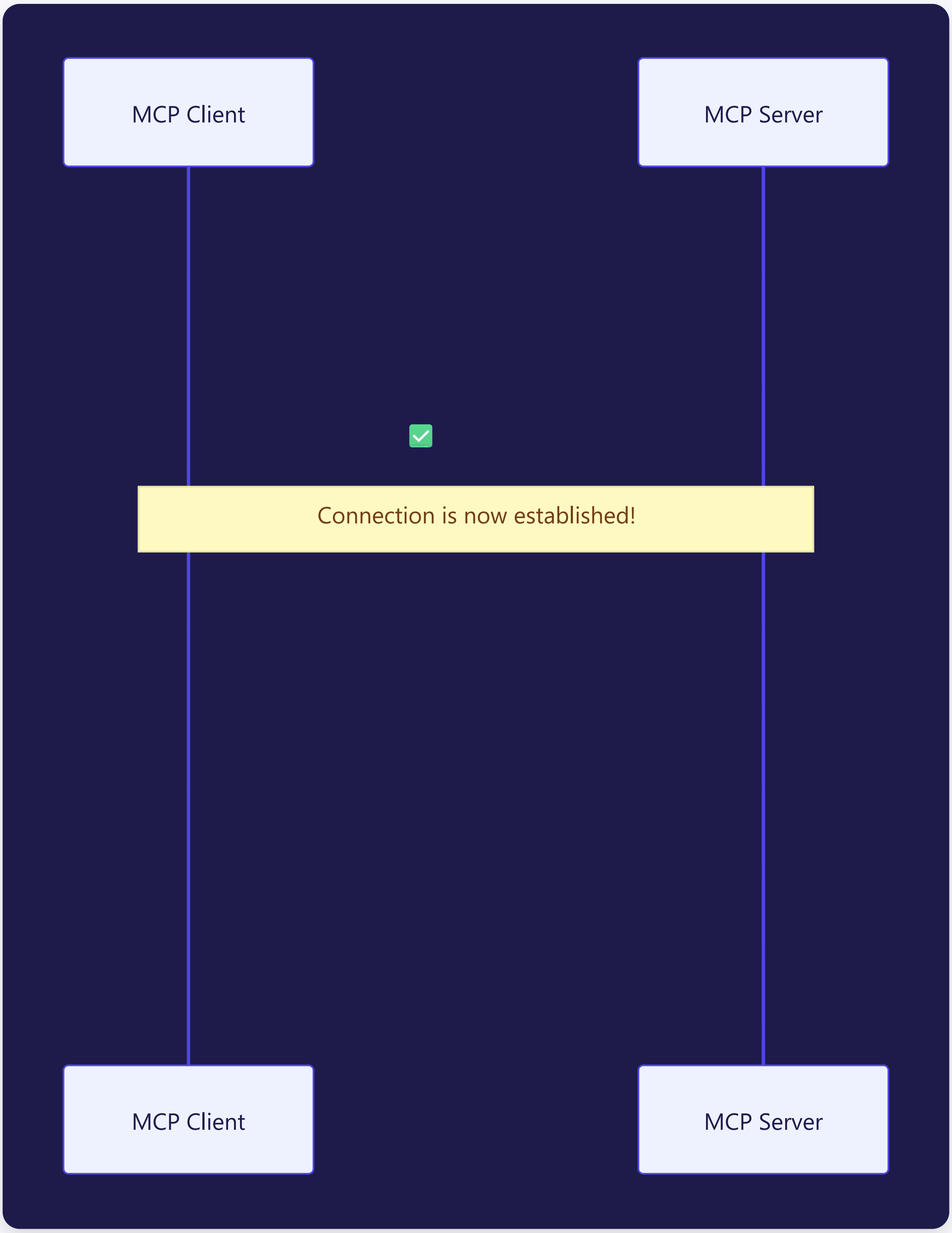
4. How a Conversation Actually Works (Step-by-Step Flow)

Here's what happens when you ask Claude "*Create a file called hello.txt*":



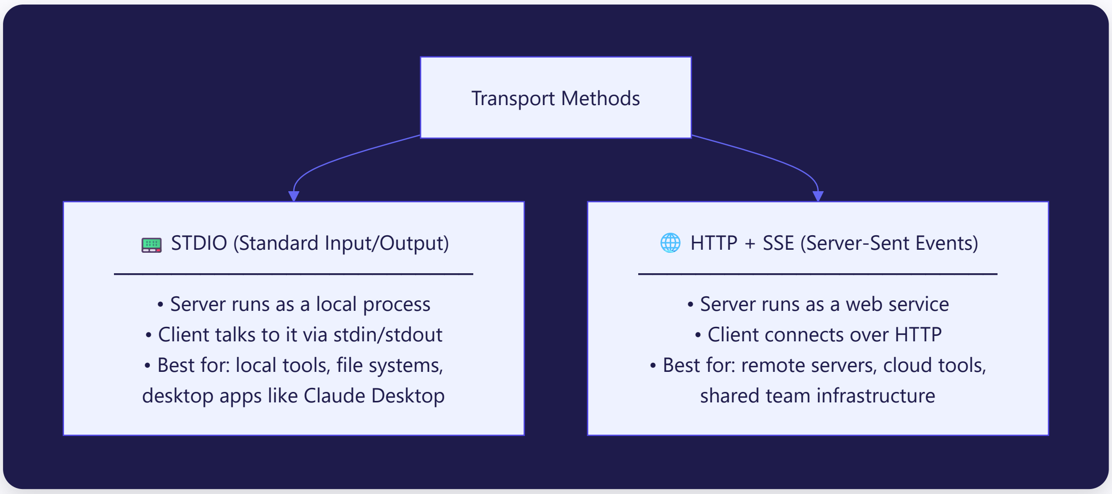
5. The MCP Handshake — How Client & Server Connect

When an MCP client first connects to a server, they do a **handshake** to discover capabilities:



6. MCP Communication — The Transport Layer

MCP servers can communicate with clients over **2 transport types**:



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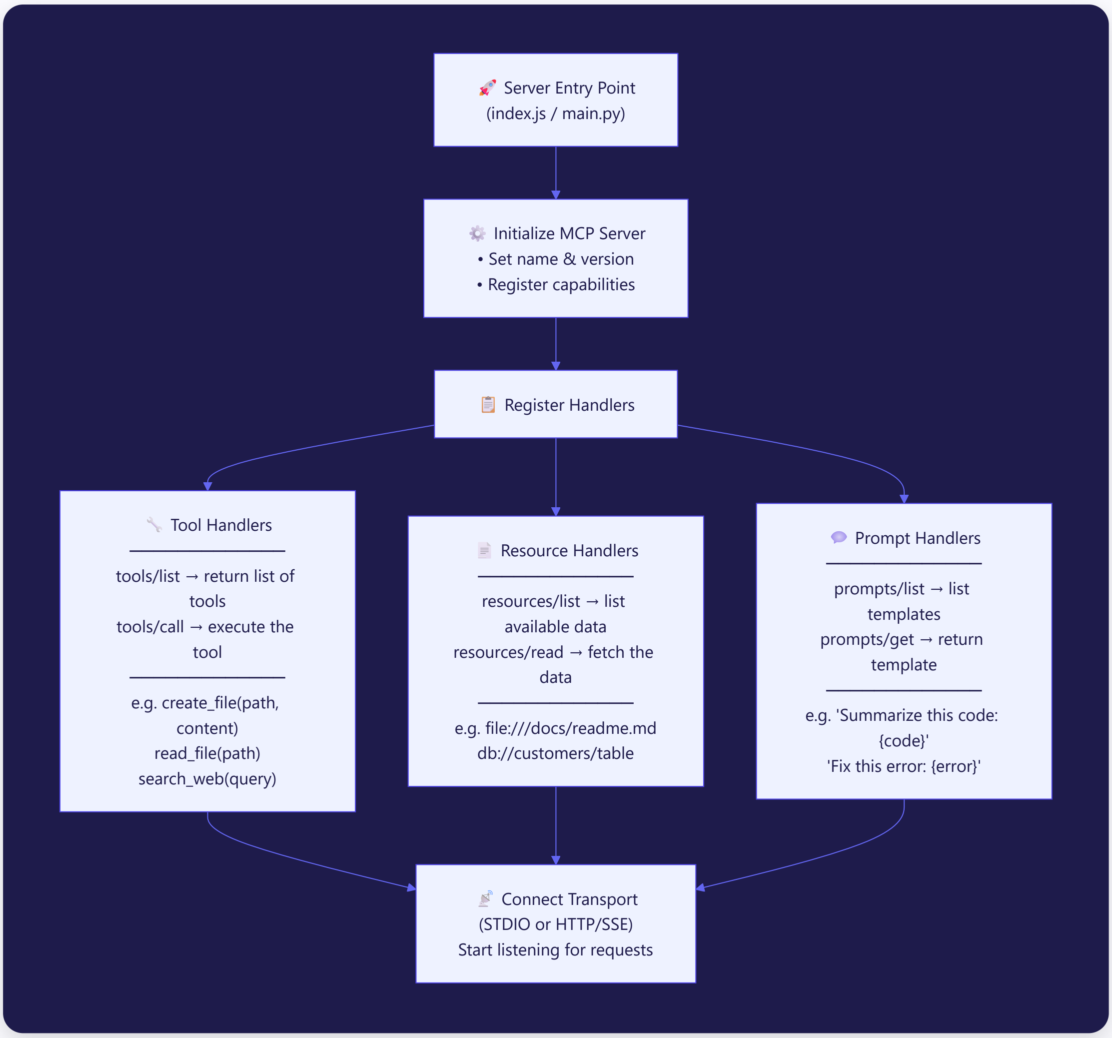
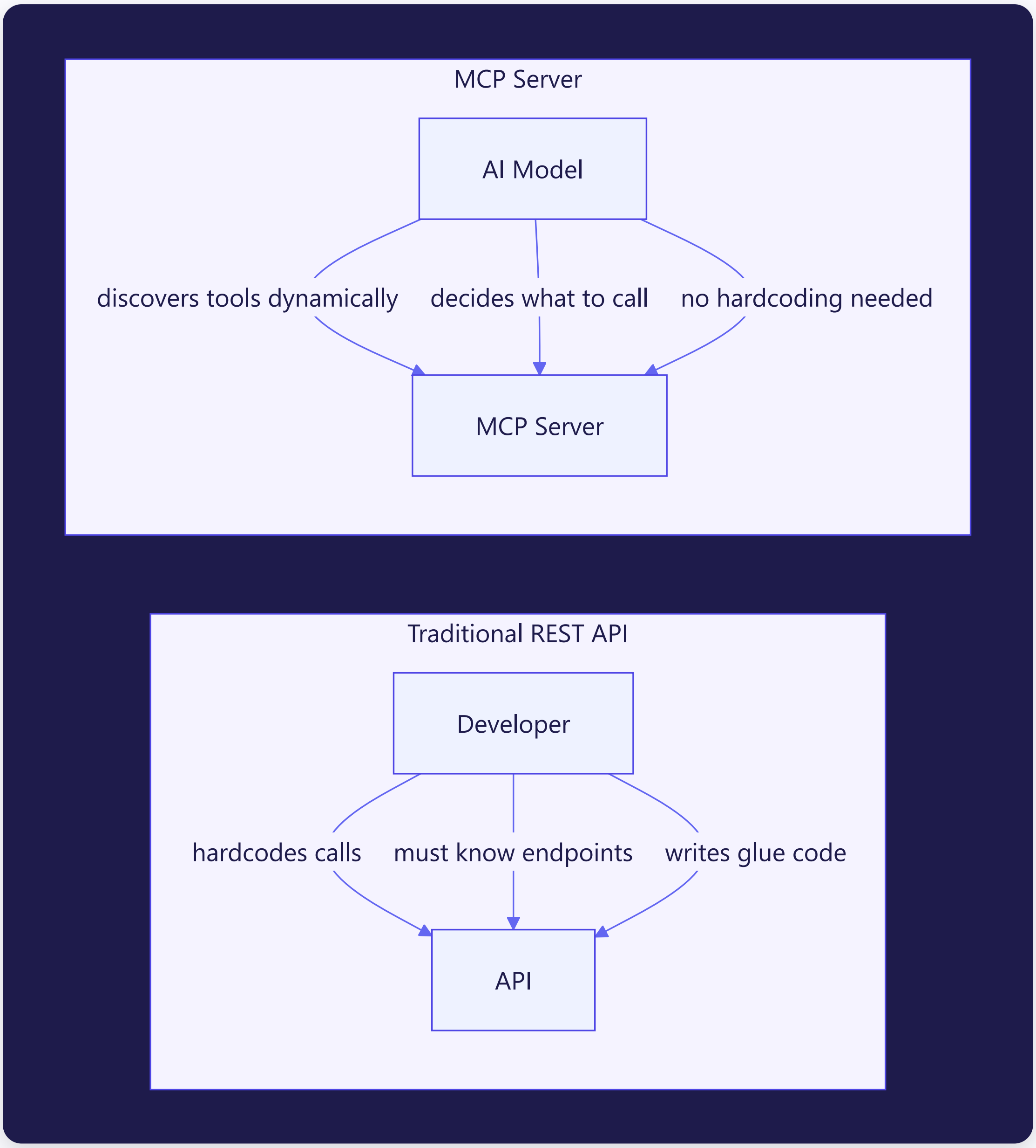
7. What Does an MCP Server Actually Look Like?

Here's a simplified mental model of an MCP server's code structure:



8. A Real-World Example — Claude + GitHub MCP Server

9. MCP vs Traditional APIs — Key Differences



Feature	REST API	MCP Server
Who decides what to call?	Developer (hardcoded)	AI Model (dynamic)
Discovery	Manual docs reading	Auto via `tools/list`
Context awareness	None	Full conversation context
Error handling	Manual	Built into protocol
AI-native?	❌	✅

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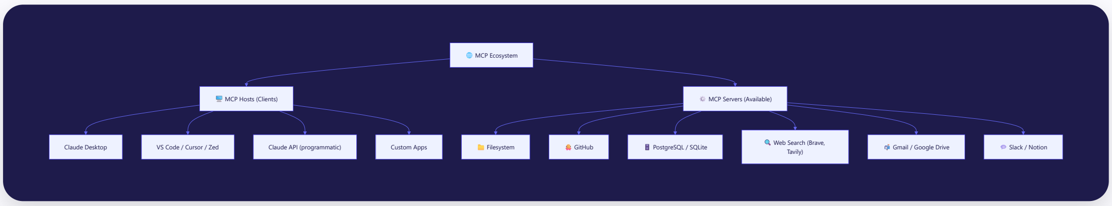
10. The Ecosystem Today



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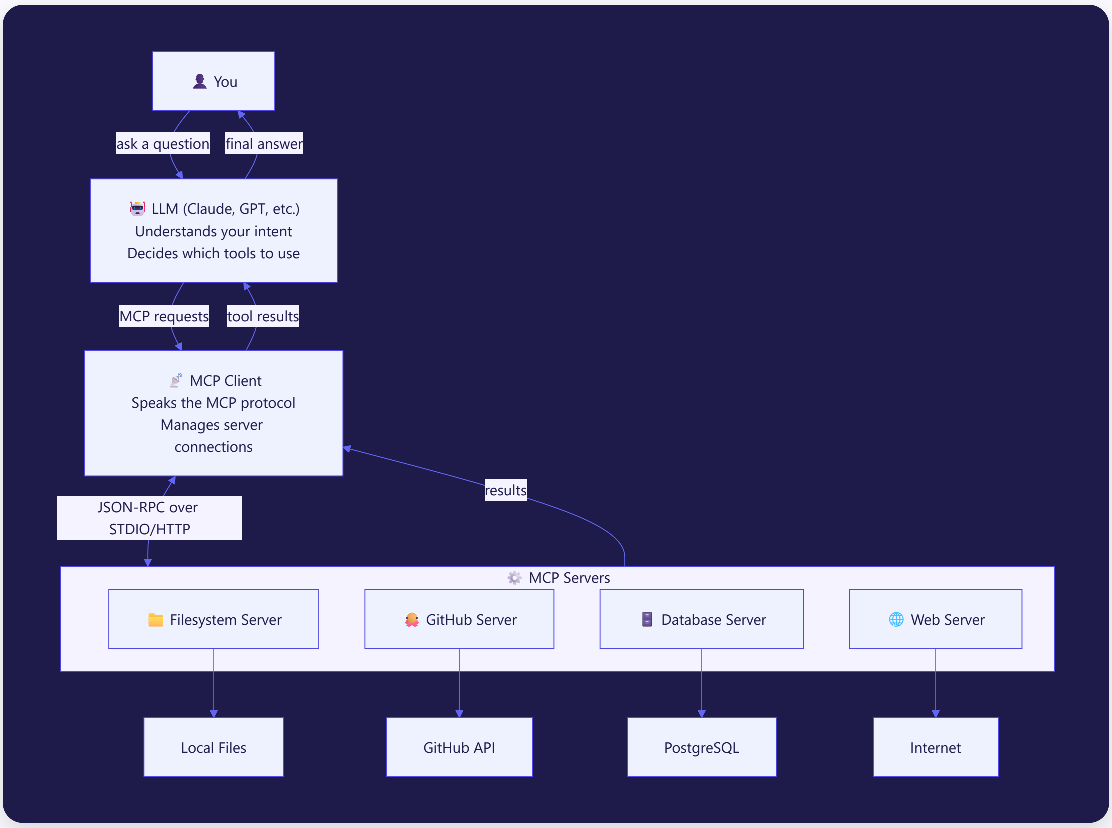
11. Summary — MCP in One Diagram



Key Takeaways 🎯

- **MCP = USB-C for AI** — one standard plug for all tools
- An **MCP Server** wraps a tool/data source and exposes it as **Tools**, **Resources**, or **Prompts**
- The **MCP Client** (inside your app) discovers and calls these servers on behalf of the AI
- Communication uses **JSON-RPC 2.0** messages over **STDIO** (local) or **HTTP+SSE** (remote)
- The AI **dynamically decides** what to call — no hardcoding needed

> *MCP was open-sourced by Anthropic in November 2024 and is now a rapidly growing standard adopted by Microsoft, Google, and hundreds of community contributors.*